

Can use Laws of Logs

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Introduction

Logarithmic functions are the inverses of exponential functions. They have several important properties that make them useful in mathematics, science, and engineering. This document will explore these properties in detail.

Definition of Logarithm

The logarithm of a number is the exponent to which the base must be raised to produce that number. Formally, for $b > 0$, $b \neq 1$,

$$\log_b(x) = y \quad \text{if and only if} \quad b^y = x$$

where: b is the base, x is the argument, y is the logarithm.

Properties of Logarithms

Logarithmic functions have several key properties that are useful in simplifying expressions and solving equations. In the following 7 properties will be discussed.

<i>Log of one</i>	$\log_b(1) = 0, \quad b > 0$
<i>Log of base</i>	$\log_b(b) = 1, \quad b > 0$
<i>Power rule</i>	$\log_b(x^y) = y \log_b(x)$
<i>Product rule</i>	$\log_b(xy) = \log_b(x) + \log_b(y)$
<i>Quotient rule</i>	$\log_b\left(\frac{x}{y}\right) = \log_b(x) - \log_b(y)$
<i>Change of base</i>	$\log_b(x) = \frac{\log_k(x)}{\log_k(b)}$
<i>Inverse rule</i>	$\log_b(y) = x \Rightarrow y = b^x$

Logarithm of 1

The logarithm of 1 to any base is 0.

$$\log_b(1) = 0 \quad \text{for any base } b > 0$$

Example: $\log_5(1) = 0$

Logarithm of the Base

The logarithm of the base to itself is 1.

$$\log_b(b) = 1$$

Example: $\log_7(7) = 1$

Power Rule

The logarithm of a power is the exponent times the logarithm of the base.

$$\log_b(x^y) = y \log_b(x)$$

Example: Simplify $\log_5(25)$

$$\begin{aligned}\log_5(25) &= \log_5(5^2) \\ &= 2 \log_5(5) \quad [\text{Power rule}] \\ &= 2 \cdot 1 \quad [\text{Log of base}] \\ &= 2\end{aligned}$$

Product Rule

The logarithm of a product is the sum of the logarithms of the factors.

$$\log_b(xy) = \log_b(x) + \log_b(y)$$

Example: Simplify $\log_2(8 \cdot 4)$:

$$\begin{aligned}\log_2(8 \cdot 4) &= \log_2(8) + \log_2(4) && \text{[Product rule]} \\ &= \log_2(2^3) + \log_2(2^2) \\ &= 3 \log_2(2) + 2 \log_2(2) && \text{[Power rule]} \\ &= 3 \cdot 1 + 2 \cdot 1 && \text{[Log of base]} \\ &= 3 + 2 \\ &= 5\end{aligned}$$

Quotient Rule

The logarithm of a quotient is the difference of the logarithms.

$$\log_b \left(\frac{x}{y} \right) = \log_b(x) - \log_b(y)$$

Example: Simplify $\log_5 \left(\frac{100}{7} \right)$:

$$\begin{aligned} \log_5 \left(\frac{100}{7} \right) &= \log_5(100) - \log_5(7) && \text{[Quotient rule]} \\ &= \log_5(10^2) - \log_5(7) \\ &= 2 \log_5(10) - \log_5(7) && \text{[Power rule]} \\ &= 2 \log_5(5 \cdot 2) - \log_5(7) \\ &= 2(\log_5(5) + \log_5(2)) - \log_5(7) && \text{[Product rule]} \\ &= 2(1 + \log_5(2)) - \log_5(7) && \text{[Log of base]} \\ &= 2 + 2 \log_5(2) - \log_5(7) \end{aligned}$$

Do it yourself: Simplify $\log_{10} \left(\frac{100}{10} \right)$

Change of Base Formula

The logarithm can be converted to a different base using the change of base formula.

$$\log_b(x) = \frac{\log_k(x)}{\log_k(b)}$$

Example: Convert $\log_{100}(3)$ to base 10:

$$\begin{aligned}\log_{100}(3) &= \frac{\log_{10}(3)}{\log_{10}(100)} && \text{[Change of base formula]} \\ &= \frac{\log_{10}(3)}{\log_{10}(10^2)} \\ &= \frac{\log_{10}(3)}{2 \log_{10}(10)} && \text{[Power rule]} \\ &= \frac{\log_{10}(3)}{2 \cdot 1} && \text{[Log of base]} \\ &= \frac{1}{2} \log_{10}(3)\end{aligned}$$

Inverse Property

The logarithm function is the inverse of the exponential function.

$$\log_b(y) = x \Rightarrow y = b^x$$

Example: Solve for x :

$$\log_2(5x) = 3 \quad \Rightarrow 5x = 2^3 \quad \Rightarrow x = \frac{8}{5}$$

From the property we also can write,

$$\log_b(b^x) = x \quad \text{and} \quad b^{\log_b(x)} = x$$

Example: $\log_2(2^4) = 4$

Example: $2^{\log_2(3)} = 3$

Applications of Logarithmic Properties

These properties are widely used in simplifying logarithmic expressions, solving logarithmic equations, and in applications such as compound interest, population growth, and information theory.

Examples

1. Simplifying Logarithmic Expressions:

$$\begin{aligned}\log_3 \left(\frac{27}{25} \right) + 2 \log_3(15) \\&= \log_3(27) - \log_3(25) + 2 \log_3(3 \cdot 5) \quad [\text{Quotient rule}] \\&= \log_3(3^3) - \log_3(5^2) + 2 [\log_3(3) + \log_3(5)] \quad [\text{Product rules}] \\&= 3 \log_3(3) - 2 \log_3(5) + 2 \log_3(3) + 2 \log_3(5) \quad [\text{Power rules}] \\&= 3 \cdot 1 - \cancel{2 \log_3(5)} + 2 \cdot 1 + \cancel{2 \log_3(5)} \quad [\text{Log of base}] \\&= 5\end{aligned}$$

2. Solving Logarithmic Equations: $\log_2(x) + \log_2(8) = 5$

$$\log_2(x) + \log_2(8) = 5 \Rightarrow \log_2(8x) = 5 \Rightarrow 8x = 2^5 = 32 \Rightarrow x = 4$$

Another way,

[Inverse property]

$$\log_2(x) + \log_2(2^3) = 5 \Rightarrow \log_2(x) + 3 = 5 \Rightarrow \log_2(x) = 2 \Rightarrow x = 2^2 \Rightarrow x = 4$$

Natural Logarithm Function

The natural logarithm function, denoted as $\ln(x)$, is one of the most important functions in mathematics. It is the inverse of the exponential function with base e , where $e \approx 2.71828$. The natural logarithm has a wide range of applications in calculus, physics, engineering, and other fields.

Natural logarithm is basically log of base e

$$\log_e(x) = \ln(x)$$

Definition

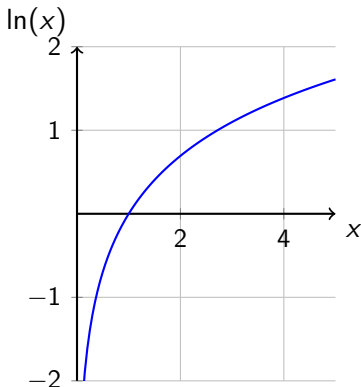
The natural logarithm of a number x is defined as the power to which the base e must be raised to obtain x . Mathematically, this is expressed as:

$$\ln(x) = y \quad \text{if and only if} \quad e^y = x$$

where e is the base of the natural logarithm, approximately equal to 2.71828.

Properties

- ▶ The domain of $\ln(x)$ is $x > 0$.
- ▶ The range of $\ln(x)$ is all real numbers, $-\infty < \ln(x) < \infty$.
- ▶ As x approaches 0 from the right, $\ln(x)$ approaches $-\infty$.
- ▶ As x increases, $\ln(x)$ increases, but at a decreasing rate.
- ▶ The function passes through the point $(1, 0)$, since $\ln(1) = 0$.



Logarithm of e

The natural logarithm of e is 1:

$$\log_e(e) = \ln(e) = 1$$

This is because $e^1 = e$. Also,

$$e^{\ln(x)} = x$$

Applications of the Natural Logarithm

The natural logarithm function is used in many applications, including:

- ▶ Calculating compound interest in finance.
- ▶ Modeling exponential growth and decay in biology and chemistry.
- ▶ Solving differential equations in physics and engineering.
- ▶ Analyzing data in statistics and economics.